OFF-GRID VEHICLE TRACKER

USER MANUAL, FOR EVERYDAY DRIVERS

Two radios, two dashboard displays, and nothing else. Each vehicle always knows where the other one is: distance, direction, speed, and short messages, with no cell phone, no cell tower, and no internet anywhere in the system.



GROUP 9 · CAPSTONE PROJECT LAB · TEXAS TECH UNIVERSITY · SPONSOR: DR. RICHARD GALE · REV 1 · 2026-09-30

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1. What this system is

This kit lets two vehicles keep track of each other anywhere, including places with no phone signal at all: deserts, mountains, long rural highways, another country. Each vehicle carries a small green **radio** and a dashboard **display**. The radios talk directly to each other over the air, like walkie-talkies that never need pressing. Every few seconds, each radio quietly tells the other one "here is where I am." Your display turns that into an arrow, a distance, and plain words.

> **In plain words:** it works like a compass that points at your friend instead of north. No accounts, no apps, no monthly fees, no coverage maps. If both boxes are switched on and charged, it works.

The two radios have names, written on their cases: **SF** and **RG**. Your display shows your partner's name at the top, so you always know who the arrow is pointing at.

2. What is in the kit

Item	What it does	What to check
Radio "SF" (green case)	Vehicle 1's transmitter. Finds its own position from GPS satellites and shares it over the air.	Antenna screwed on, battery charged, small screen lights up.
Radio "RG" (green case)	Vehicle 2's transmitter. Same job as SF.	Same checks.
Display 1	The dashboard screen that rides with SF. Shows where RG is.	Powers on, shows "link OK" at the bottom once its radio is on.
Display 2	The dashboard screen that rides with RG. Shows where SF is.	Same checks.
Antennas (black sticks)	The radios' voice and ears. Nothing works without them.	Hand-tight on the round connector. Never transmit without one.
USB-C cables and chargers	Charging for radios and power for displays.	Charge radios after every trip.

Important pairing: each display belongs to one radio. Display 1 works with SF. Display 2 works with RG. Keep each pair together in the same vehicle.

3. Before every trip: the two-minute start-up

PRE-FLIGHT CHECKLIST

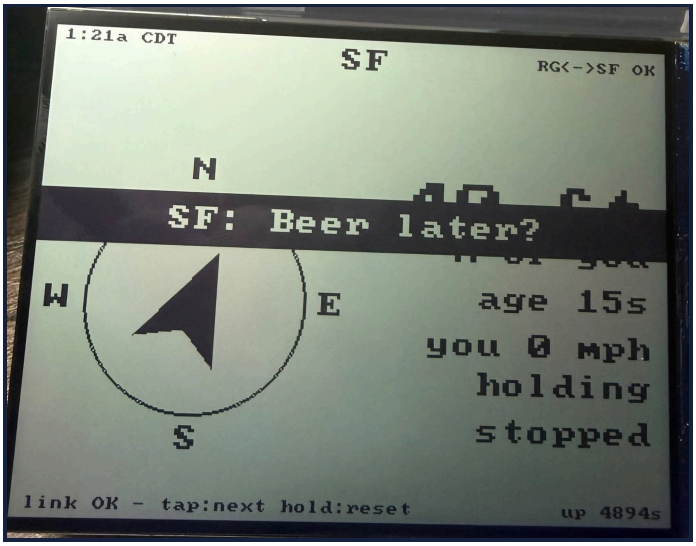
1. **Charge check.** Both radios above 20 percent. A radio that dies mid-trip can forget its settings, and then the two radios stop understanding each other.
2. **Antennas on.** Both radios, hand-tight.
3. **The magic number.** On each radio, tap the button to reach the screen labeled "LoRa" and confirm it reads **912.875**. Both radios must show this exact number. It is the private channel the two radios share. If either shows a different number, see Section 8.
4. **Power on together.** In each vehicle, switch on the radio and the display at the same time, like turning the ignition key. The display finds its radio during the radio's first minute. The display footer then reads `link OK`.
5. **Sky view.** Position updates need GPS. Give each radio a view of the sky (dashboard, window mount). Indoors, in a garage, positions will honestly show as missing until you drive out.

Within a minute or two of both vehicles being on, each display shows the partner's name, an arrow, and a distance. You are ready to drive.

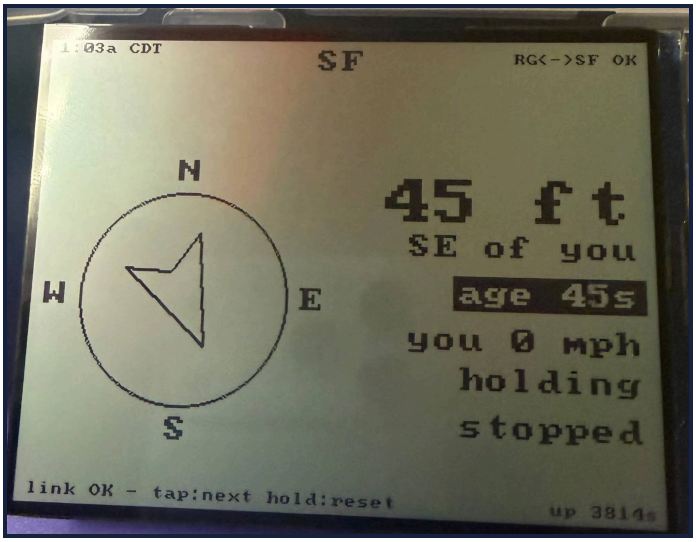
4. Reading your screen

The display has three pages. A short press of its button (`tap: next`) moves to the next page.

Page 1: The compass (your driving page)



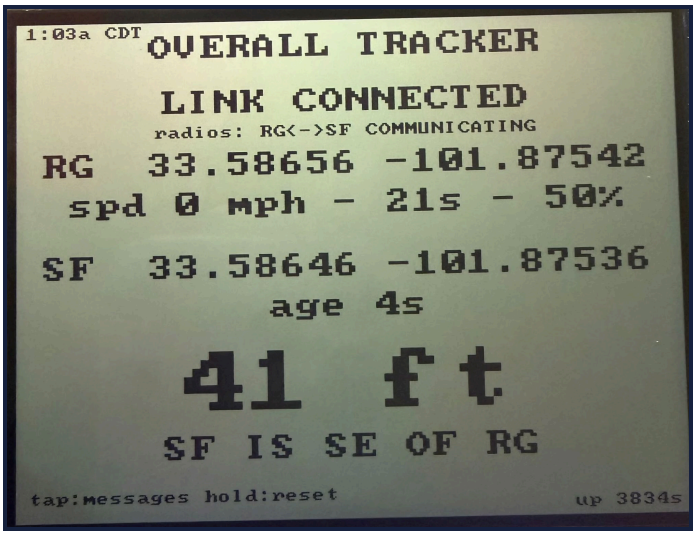
compass page, with a message arriving



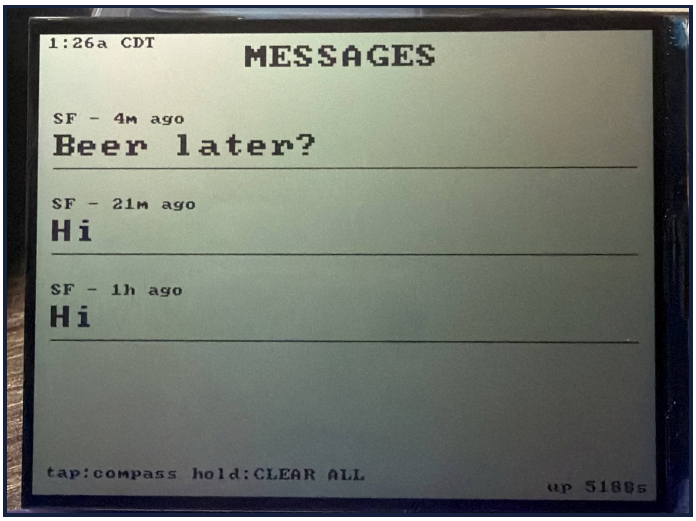
compass page, data getting old (hollow arrow)

What you see	What it means
Name at the top (SF or RG)	Who the arrow points at: your partner vehicle.
The dark arrow in the circle	The direction to your partner. Solid arrow: the position is fresh. Hollow outline arrow: the position is an estimate because no update has arrived for half a minute; trust it less.
Big number (feet or miles)	Straight-line distance to your partner.
"SE of you" or "at 2 o'clock"	The direction in words. While you drive, it uses clock positions relative to your hood. While you are stopped, it uses compass words.
"age 15s"	How old the partner's information is, in seconds. Small is good. If it turns into a dark highlighted box, the data is getting stale.
"you 0 mph"	Your own speed.
"closing 12 mph" / "apart 8 mph" / "holding"	Whether the gap between vehicles is shrinking, growing, or steady.
"RG<->SF OK" (top right)	The radios are actively hearing each other. If this tag disappears, they have gone quiet.
"link OK" (bottom)	The display is connected to its own radio in your vehicle.

Page 2: The status page (the numbers page)



status page: positions, ages, battery, distance



messages page with history

This page lists both vehicles by name with their exact coordinates, speeds, data ages, and each radio's battery percentage. The line "radios: RG<->SF COMMUNICATING" confirms the two radios are in contact. The big number is the distance between the two vehicles, and the sentence under it, such as "SF IS SE OF RG", says the same thing in words.

Page 3: Messages

The last few messages you received, newest first, each with the sender's name and how long ago it arrived. Holding the button on this page (`hold: CLEAR ALL`) erases the list. Holding the button on any other page resets the tracking data instead, which is harmless: fresh data rebuilds within seconds.

5. Sending and receiving messages

Messages are **sent from the radio** (the small green box) and **appear on the partner's display**. The radio stores 21 ready-made messages so nobody types while driving. Press the radio's button to open the list, keep pressing to scroll, and long-press to send the highlighted one.

Group	Messages
Quick replies	OK · Yes · No · Hello
Checking in	How are you? · I am good
Safety	Help! · Are you hurt?
Position	Send position
Driving	On my way · Wait there · I am here · Follow me · Slow down · Stop · Turn left · Turn right · Turn back
Social and status	Beer later? · Low battery · Going home

When a message arrives, it appears as a dark banner across the partner's display for about eight seconds, for example **"SF: Beer later?"**, and is also saved on the Messages page. Only the receiving vehicle gets the banner. Your own display stays quiet when you send, which is how you know the banner you see is always news from the other car.

> **In plain words:** every question in the list has its answer in the list. "Are you hurt?" can be answered with "I am good" or "Help!". "Beer later?" can be answered with "Yes". The list is a conversation kit.

6. Asking "where are you right now?"

Positions update automatically: every ten seconds while a vehicle is moving, every minute while it is parked. If you want the partner's position immediately, press the **BOOT button on your display**. Your radio asks the partner's radio directly, and the partner's radio answers within a few seconds. You will see the partner's "age" number drop to nearly zero. This is useful when the partner has been parked and quiet and you want to confirm where they are before setting off.

7. Warnings, and what to do about them

On the screen	What it means	What to do
RADIO LINK LOST	The display cannot reach its own radio in your vehicle.	Check the radio is on and within a few feet. Power the radio and display off, then on together.
SF DATA STALE (or RG)	No position from the partner for over two minutes.	Usually distance or terrain. The arrow hollows and ages count up honestly. It heals itself when you get closer.
RG BATTERY 15% (or SF)	The partner's radio battery is nearly empty.	Tell them to plug it in. A dead radio disappears from your screen entirely.
Hollow arrow, highlighted age	The position shown is an estimate, not a fresh measurement.	Treat the arrow as approximate until it turns solid again.
CHANGE COURSE >>	You have been driving away from your partner for a while.	Follow the chevrons if your goal is to rejoin them.

8. When something seems wrong

"The other vehicle is not on my screen at all."

1. Is the partner's radio switched on, with its screen lit? A dark screen means a dead battery.
2. Does each radio's LoRa screen read **912.875**? If one shows a different number, that radio lost its settings, usually after its battery ran completely flat. It needs the recovery procedure in the technical manual, a five-minute fix with a laptop.
3. Power-cycle both radios (off, count to five, on). They announce themselves on boot and usually reappear on each other's screens within a minute.

"The distance looks wrong."

Distance is measured between two GPS positions. Indoors or parked by a building, each GPS can be off by a few dozen feet, so two parked vehicles may read 40 feet apart while sitting side by side. This tightens up outdoors and while moving. If distance is wildly wrong (miles apart while parked together), one radio is holding an old position: press the display's BOOT button to request a fresh one.

"The names show letters and numbers instead of SF or RG."

The display learns names from the radios' introductions. After a reset it may briefly show a serial-style name like "bb5c". It corrects itself automatically; power-cycling the partner radio speeds it up.

9. Taking care of the equipment

- **Charge after every trip.** A radio left switched on runs its battery flat in one to two days.
- **Switch radios off when stored.** Off, they hold charge for months. A battery that runs completely flat can erase the radio's settings.
- **Never transmit without an antenna.** Operating a radio with the antenna removed can damage it permanently.
- **The printed cases** protect the screens and keep every port reachable. If a case is ever opened, use only the original short screws: longer screws press on the glass.
- **Sun and heat.** The dashboard display is sunlight-readable by design and produces almost no heat. The radios prefer shade like any battery device.

10. One-page quick reference

I want to...	Do this
Start the system	Antennas on, radios charged, power radio + display together in each vehicle, confirm "link OK".
See where my partner is	Compass page: arrow, distance, direction in words, data age.
See exact numbers and batteries	Tap to the status page.
Send a message	On the radio: press to open the list, press to scroll, long-press to send.
Read past messages	Tap to the Messages page on the display.
Erase past messages	Hold the display button while on the Messages page.
Get partner's position right now	Press BOOT on the display.
Fix "partner missing"	Check partner power, check both radios read 912.875, power-cycle both.
Store the kit	Charge everything, switch radios off.